

Towhid Ahmed

Generative AI Engineer · Open to remote — US/EU time zones

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Professional Summary

Generative AI Engineer specializing in **production-grade LLM applications, multi-agent systems**, and advanced Retrieval Augmented Generation (RAG). Built **5 AI applications** using **LangChain, LangGraph, FastAPI**, and **modern LLMs**, achieving up to **35% hallucination reduction**, **25% retrieval** relevance improvement, and **98% test coverage**. Experienced in building scalable AI backends, **RAG pipelines**, and **LLMOps** workflows with Docker and **LangSmith**.

Technical Skills

Generative AI & Agentic Systems: LangChain, LangGraph, Multi-Agent Systems, Tool Calling, Prompt Engineering, Advanced RAG, Multimodal RAG, RAG Pipeline, Retrieval Strategies, Cross-Encoder Re-ranking, Vector Databases (Pinecone, Weaviate, ChromaDB)

LLMOps & Deployment: LangSmith, LLM Guardrails, FastAPI, Docker, Streamlit, CI/CD (GitHub Actions), Render

LLMs & AI Frameworks: Llama, OpenAI API, Groq, Hugging Face Transformers, Sentence Transformers

Machine Learning & Deep Learning: Scikit-learn, Feature Engineering, Supervised & Unsupervised Learning

Programming & Tools: Python, PostgreSQL, SQL, Git, GitHub, Pandas, NumPy, Java

Software Engineering: Data Structures & Algorithms (**Solved 400+ problems**), OOP, LLM System Design

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | June 2026 - July 2026

- Built **3-agent LangGraph** workflow with Pinecone semantic search and citation-based RAG **reducing hallucinations**
- **Tech:** FastAPI, LangChain, OpenAI Embeddings, PostgreSQL, Streamlit; Designed and implemented **7 REST APIs**
- Production system processing **1000+ documents** with **error handling, logging**, and **session management**

AI Sales & Customer Support Assistant | Jan 2026 - Feb 2026

- Architected **multi-agent LangGraph** pipeline (Intent, Retrieval, Response) with Pinecone RAG for **automated sales/support**
- **Tech:** FastAPI backend with **6+ REST APIs**, Streamlit UI with knowledge base **uploader, session memory**
- **Optimized:** Free-tier Groq LLM, local HuggingFace embeddings, Pinecone serverless; Docker deployment for e-commerce/SaaS

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, HuggingFace, Tavily GitHub | Demo

- Designed **4-agent LangGraph** workflow routing queries between **RAG** and **web search** based on intent and complexity
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**
- Integrated **Tavily** with **DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25%** and **reducing hallucinations by 35%**
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**
- **Tech:** Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker; 76-unit test coverage

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built **clinical AI** using **4-agent LangGraph** pipeline (Router+Safety+Retrieval+Answer) with **RAG** and **web search**
- Implemented **multi-layer safety system** (input filtering, retrieval validation, output sanitization) with JWT-secured APIs
- Achieved **98/98** passing tests across **guardrails, RAG evaluation**, and **API workflows**

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology, Bangladesh

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMOps — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward